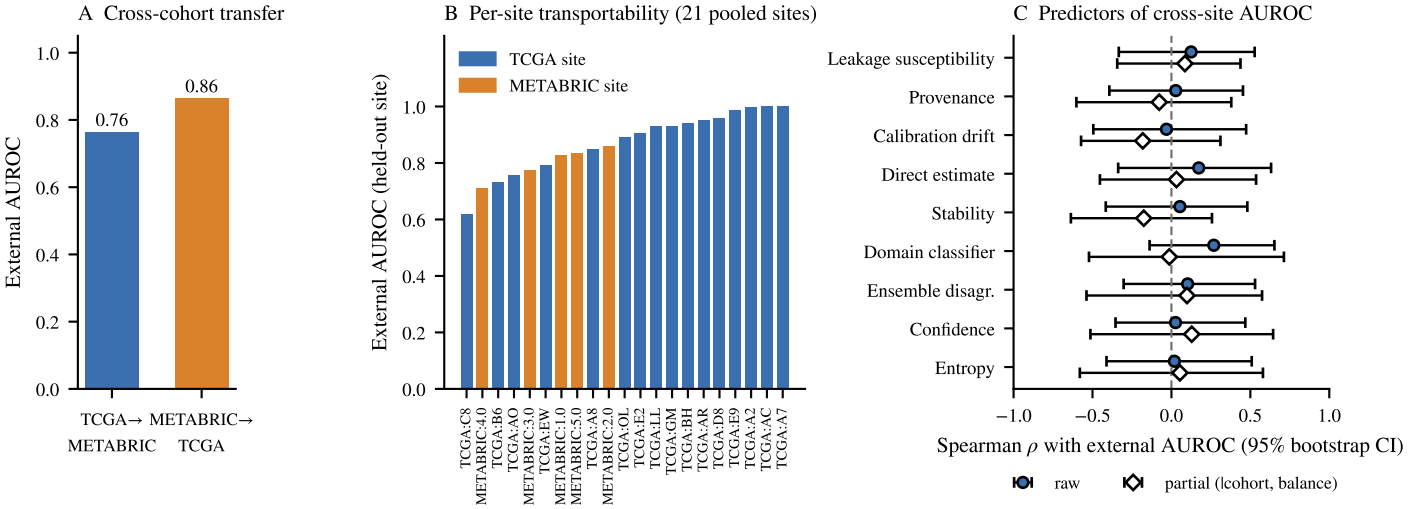
A Cross-cohort transfer
External AUROC
1.0
0.8
0.6
0.4
0.2
0.0
0.76
0.86
TCGA→ METABRIC
METABRIC→ TCGA
B Per-site transportability (21 pooled sites)
TCGA site
METABRIC site
External AUROC (held-out site)
1.0
0.8
0.6
0.4
0.2
0.0
TCGA:C8
METABRIC:4.0
TCGA:B6
TCGA:AO
METABRIC:3.0
TCGA:EW
METABRIC:1.0
METABRIC:5.0
TCGA:A8
METABRIC:2.0
TCGA:OL
TCGA:E2
TCGA:LL
TCGA:GM
TCGA:BH
TCGA:AR
TCGA:D8
TCGA:E9
TCGA:A2
TCGA:AC
TCGA:A7
C Predictors of cross-site AUROC
Leakage susceptibility
Provenance
Calibration drift
Direct estimate
Stability
Domain classifier
Ensemble disagr.
Confidence
Entropy
−1.0
−0.5
0.0
0.5
1.0
Spearman ρ with external AUROC (95% bootstrap CI)
raw
partial (lcohort, balance)